Investment Decision Theory and Application- ECO4102Z/ECO4122Z

Section Overview

* in test + exam

- Investment objectives for individual and institutional investors
- Portfolio Management Process
- Investor Constraints
- Investment Policy Statement: Individual vs. Institutional
- Importance of Asset Allocation

Based on Reilly and Brown Chapter 2 - See Chapter extract on Vula

"My 35 years of experience tell me your tolerance for risk is low..."

2:00

Asset Allocation-Key Terms

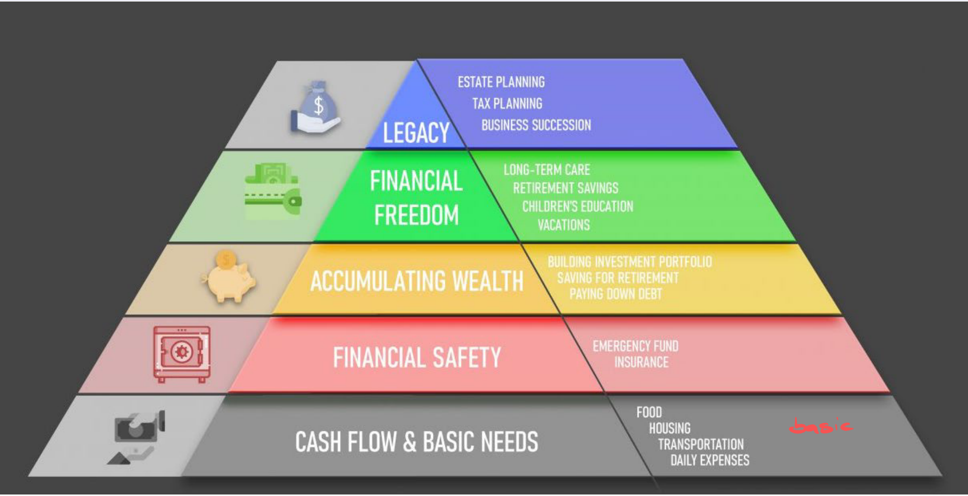
Asset Allocation

Process of deciding how to distribute an **investor's** wealth among different countries and **asset classes** for investment purposes

- Asset Class: group of securities with similar characteristics, attributes, and risk/return relationships
- Investor: investment objectives and constraints may vary depending on type of investor - NB for this topic
  - Individual investors: save and invest for a variety of reasons,
  - Institutional investors: have large investment portfolios.

Individual Investors

- Financial plans and investment needs- Maslow Hierarchy ? e.g. buy property , education etc
- Need buffer for unexpected events, which may include
  - **Life Insurance:** providing death benefits and, possibly, additional cashvalues
    - Term life and whole life insurance
    - Universal and variable life insurance
  - **Non-life Insurance**
    - Health insurance and disability insurance
    - Auto-mobile insurance and Home/rental insurance
  - **Cash Reserve**
    - To meet emergency needs- job loss, unforeseen expenses or investment opportunities
    - Should be able to cover significant period- (six months?)
  - In most countries, special accounts allow citizens to invest for retirement and to defer any taxes on investment income and gains until the funds are withdrawn.


LEGACY
ESTATE PLANNING
TAX PLANNING
BUSINESS SUCCESSION
FINANCIAL FREEDOM
LONG-TERM CARE
RETIREMENT SAVINGS
CHILDREN'S EDUCATION
VACATIONS
ACCUMULATING WEALTH
BUILDING INVESTMENT PORTFOLIO
SAVING FOR RETIREMENT
PAYING DOWN DEBT
FINANCIAL SAFETY
EMERGENCY FUND
INSURANCE
CASH FLOW & BASIC NEEDS
FOOD
HOUSING
TRANSPORTATION
DAILY EXPENSES
basic

9:25

Institutional Investors

- Professional investors who provide investment management services for a fee. Some are employed directly by wealthy individual investors. Most would, either pool many individual investor funds and manage them or serve institutional investors.
  - **Personal trust** e.g family trust
  - **Mutual funds** - manage pool of individual investors' money. e.g unit trusts cheaper
  - **Pension funds**- of two basic types; defined contribution and defined benefit. focus: where they put money
  - **Life insurance companies**- investment driven by the need to hedge their liabilities, derived from the policies they write.
  - **Non-Life-Insurance Companies**- e.g. property and casualty insurers. Need to cover claims in excess of premiums. Tend to be conservative in their attitude toward risk. Typically use bonds with various maturities to hedge their liabilities.
  - **Investment Banks**-seek to match risk of assets to liabilities and thus earning profit on spread between lending and borrowing rates.

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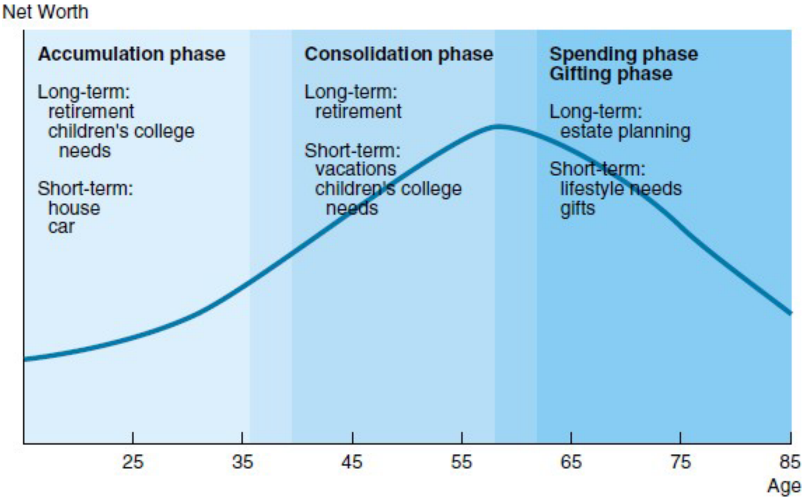
Life-cycle view of Individual Investor Behaviour



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Life Cycle Investment Goals

1. **Near-term, high-priority goals**
   - High emotional importance Short
   - time horizon,
   - Therefore usually achieved through high-risk investments
2. **Long-term, high-priority goals**
   - Long-term nature, and can be achieved through higher-risk investments
3. **Lower-priority goals**
   - Non-critical, e.g. home decoration, holiday vacation, etc

In designing an investment portfolio for an investor the above should be considered.

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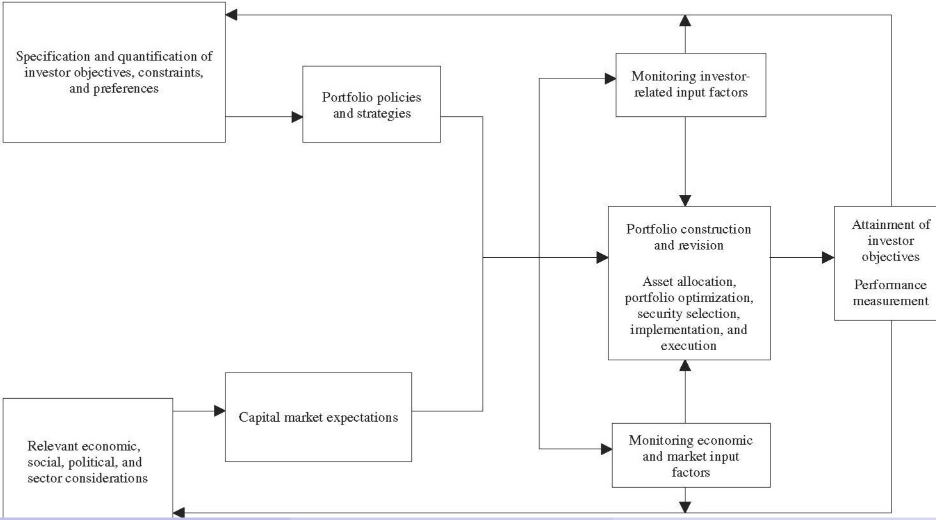
Portfolio Construction, Monitoring, and Revision Process



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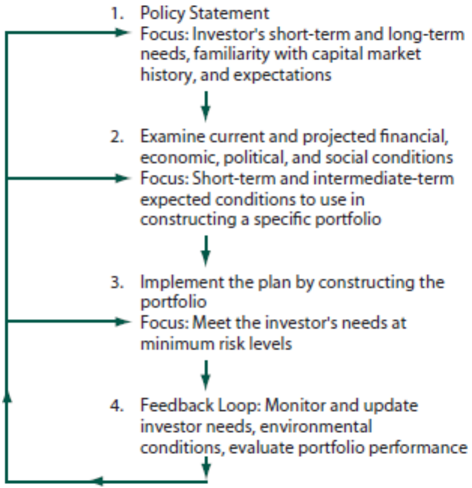
Portfolio Construction, Monitoring, and Revision Process



Portfolio Construction, Monitoring, and Revision Process

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Why Investment Policy Statement (IPS)?

- An investment policy statement sets the framework for all investment decisions that follow
- Begins with profile analysis of investor’s current and future financial situations and discussion of investment objectives and constraints
- Helps portfolio manager to understand the need of the investor
- Guides decision-making process- portfolio management, aids navigation through unexpected market fluctuations and changes in economic conditions.
- Understand investor’s needs and articulate realistic investment objectives and constraints
  - Real risks of an adverse financial outcome, and likely emotional reactions
  - Investor’s understanding of investments and financial markets
  - Capital or income sources-significance of portfolio to overall financial position.
  - Legal restrictions faced, if any
  - Impact of unanticipated change in portfolio value.

30:10

Why IPS?

- Sets standards for evaluating portfolio performance
  - Provides a standard in judging the portfolio manager's performance
  - Benchmark portfolio or comparison standard is used to reflect the risk and return objectives specified in the policy statement
  - Should act as a starting point for periodic portfolio review and client communication with the manager

31':55

Why IPS?

- **Other Benefits**
  - Reduces possibility of inappropriate or unethical behaviour by portfolio manager
  - Helps create seamless transition from one money manager to another without costly delays
  - Provides framework to help resolve any potential disagreements between the client and the manager
  - To satisfy regulatory requirements and fiduciary standards of care and client protection (suitability and fair dealing)

33:05

Input to the IPS

- A well-written policy should cover the following;
  - Philosophy or purpose.
  - Return/risk objectives, including
  - thresholds, Time horizon.
  - Type of plan or portfolio.
    don't cover
  - Actuarial assumptions, including clearly stated reasoning behind the return/risk objectives.
  - Cash flow needs or liquidity.
  - Strategy for meeting investment objectives.
  - Permissible and Restricted investments (financial instruments and asset classes).

38', 45

Input to the IPS

- Asset allocation ranges (weights).
- Benchmarks total portfolio and each individual component of the total portfolio (asset classes, individual managers, styles, etc.).
- Plan/portfolio responsibilities (board, investment committee, consultants,etc.).
- Portfolio and performance evaluations (standards and procedures), Benchmarks and rebalancing policies.
- Diversification (by manager, portfolio size, geography, investment characteristics, asset classes, etc.).
- Portfolio execution and trading strategy.
- Operational issues (custodial, administrative, spending policy).

Investment Objectives

- Critical element in the IPS- sets the tone for everything
- Should state portfolio's return and risk expectations
- For pension plans, return and risk objectives may be stated in absolute terms
  - e.g portfolio should return a minimum of 8% annually with a standard deviation no greater than 10% annually)

  or in relative terms
  - e.g the portfolio should have an annual return in excess of 200 basis points above the Top 40 Index with a standard deviation no greater than the Top 40 Index). Recall a basis point= 1/100th of a percent.

Example of Investment Objective

The financial objectives of the plan are based on a comprehensive evaluation of the capital markets in the context of modern portfolio theory and have been measured against the plan's current and projected financial needs. Based on this evaluation, the plan will be measured against a customized benchmark consisting of the following indexes in the proportions listed:

50% JSE Top 40 index, 10% JSE Industrial index, 10% JSE All Share Index, 30% JSE Financial index

The investment objectives for the plan are:

- To achieve a nominal rate of return for the total portfolio equal to or greater than the return of the customized benchmark.
- To achieve a real rate of return in excess of 550 basis points above inflation, measured by the consumer price index (CPI).
- To keep the total portfolio's level of risk, defined as annualized standard deviation, equal to or less than that of the customized benchmark.

Responsibilities

- Once the objectives have been clarified, it is important to clearly state who will be responsible goals accomplishment.
- Specify who is responsible for every aspect of the portfolio’s management- eg. boards, trustees, internal employees, external investment managers, consultants, legal advisers, and others.
- Can be viewed as a detailed organizational chart setting the pecking order within an organization.

40:45

Asset Allocation

- Provide guidelines on asset classes to include and a range of percentages to invest in each class.
- Specified as a range to allow investment manager to use own criteria based on their assessment of capital market trends
- Goal is to allocate asset class weights that produce a portfolio consistent with the individual investor's return objective, risk tolerance, and constraints.
- Should be executed from a taxable perspective, considering;
  (i) after-tax returns,
  (ii) the tax consequences of any shift from current portfolio allocations,
  (iii) the impact of future rebalancing,
  (iv) asset "location." (e.g clearly, nontaxable investments should not be "located" in tax-exempt accounts)
- See Government Employees Pension Fund IPS- South Africa
- Impact of Regulation 28 of the Pension Funds Act

42:00

Example of Detailed Asset Allocation Guidelines

Asset Allocation		Minimum %	Average %	Maximum %
Traditional Asset Classes		**80**	**90**	**100**
equity or bond	*Equity*	40	47	60
	South Africa	*32*	*35*	*44*
	Small-cap	6	7	9
	Mid-cap	7	8	10
	Large-cap	19	20	25
	Non-South African	*8*	*12*	*16*
	BRICS	6	9	12
	Emerging markets	2	3	4
	Fixed Income	20	30	40
	South African	*16*	*25*	*33*
	Investment grade	12	20	26
	Non-investment grade	4	5	7
	Non-South African	*4*	*5*	*7*
Property fits in both	Developed markets	2	3	4
	Emerging markets	2	2	3
Alternative Asset Classes		**0**	**10**	**20**
	Hedge Funds	0	3	10
	Private Equity	0	4	10
	Cash and Equivalents	1	3	5

money mkt securities (fixed income)

- The more detailed the asset allocation, the more efficient the overall process will be.
- It is easier to perform attribution analysis when the asset classes have been defined in greater detail.

44:20

Risk and Return

- Well-thought-out and well-written IPS will contain risk and return objectives that match the asset allocation ranges
- Risk objectives should be based on investor's ability to take risk and willingness to take risk
- Risk tolerance depends on an investor's current net worth and income expectations and age
  - More net worth allows more risk taking
  - Younger people can take more risk
- Careful analysis of client's risk tolerance should precede any discussion of return objectives
- See Merrill Lynch Profile and this one

45:20

Risk Objectives

49:20 Test: willingness + ability

1. **Risk measurement is a key issue in investments**
   - Risk may be measured in absolute terms (e.g standard deviation or variance of total return) or in relative terms with reference to various risk concepts.
   - Downside risk concepts, such as value at risk (VaR), may also be important to an investor.
2. **Investor's willingness to take risk**
   - Different for institutional versus individual investors.
3. **Investor's ability to take risk**
   - Even if an investor is eager to bear risk, practical or financial limitations often limit the amount of risk that can be prudently assumed.
4. **Risk tolerance**
   - The capacity to accept risk, It is a function of both an investor's willingness and ability to accept risk.
   - Can also be described in terms of risk aversion- degree of an investor's inability and unwillingness to take risk.

53:35

Risk Tolerance: Willingness vs Ability to take risk

- When an investor's willingness to accept risk exceeds ability to do so, ability prudently places a limit on the amount of risk the investor should assume.
- When ability exceeds willingness, the investor may fall short of the return objective because willingness would be the limiting factor.

[redacted] **Risk Tolerance**

Willingness to Take Risk	Ability to Take Risk	
	Below Average	**Above Average**
Below Average	Below-average risk tolerance	Resolution needed
Above Average	Resolution needed	Above-average risk tolerance

54:50

Question

Suppose Amanda is the CEO of a large multi-national company. Her lifestyle is relatively conservative and as a result, she has accumulated R10 million in savings and has paid off the mortgage over her property. She will reach retirement age in 15 years. She believes that "cash is king" and the financial markets are "just a gamble." Which of the following options best describes her ability and willingness to take on risk?

A. Ability: low; Willingness: high

B. Ability: high; Willingness: low

C. Ability: low; Willingness: low

D. Ability: high; Willingness: high

The correct answer is B.

The client's wealth is relatively substantial and exceeds their lifestyle requirement and financial obligations. The earnings are expected to continue for 15 years, a fairly long time horizon and as such, the ability to bear risk is high.

However, the client demonstrates a low willingness to take on investment risk perceiving the financial markets to be "a gamble." Therefore the willingness to take on risk is low.

56:55

Other Considerations in portfolio allocation

1. **Risk tolerance**
   - Common rule of thumb in asset allocation is the rule of 100
     - i.e. percentage of portfolio allocated to stocks is 100 minus investor age, remainder kept in bonds and other assets(other variations to this include 110 rule, 125 rule)
   - Need to adjust the above to reflect risk tolerance. technically should be 30%.
     - i.e. if one is 70 years old with a high risk tolerance they may allocate 40% in risky assets; can also adjust downwards as well.
2. **Portfolio rebalancing** occurs when portfolio goes out of line
   - The 5/25 rule: ∴ Transaction costs, so you don't want to always rebalance
     - The '5' means, if any large block asset of your portfolio deviates by 5%, then you rebalance it- helps determine the minimum and maximum allocation, e.g if target allocation is 20%, rebalance if it reaches 15%(25%) by buying(selling) assets.
     - The '25' is for smaller portions of the portfolio, for example, anything between 5-10%. For such rebalance when you lose or gain 25%.

Risk Objectives

5. **Specific risk objective(s)**
   - Need to specify both absolute risk and relative risk objectives.
   - In practice, investors often find that quantitative risk objectives are easier to specify in relative than in absolute terms.
     - For example,if an investor has ”lower than average risk tolerance”- this may be translated into an operational objective as follows: ”the loss in any one year should not exceed x %of portfolio value” or ”annual volatility of the portfolio is not to exceed y percent.”
6. **Investor risk allocation**
   - Relates to how some investors frame capital allocation decisions
   - May concern the portfolio as a whole or some part

1:06:50

Return Objectives

- May be stated in terms of an absolute or a relative percentage return
  - **Capital Preservation:** Minimize risk of real losses
  - younger: **Capital Appreciation:** Growth of the portfolio in real terms to meet future need

    dividends, coupons, rent
  - retirement **Current Income:** Focus is in generating income rather than capital gains
  - **Total Return:** Increase portfolio value by capital gains and by reinvesting current income with moderate risk exposure

1:02:10

Return Objectives

1. **Return measurement**
   - Usual measure is total return- may be stated as an absolute amount, eg. 10% a year, or relative to the benchmark's return, e.g. JSE top 40 plus 2% a year.
     - Need to distinguish between nominal returns and real returns. Nominal- unadjusted for inflation. Real returns -adjusted for inflation, sometimes simply called inflation-adjusted returns.
     - Also, pre-tax returns must be distinguished from post-tax returns.
2. **Level of return desired by the investor**
   - These wants or desires may be realistic or unrealistic.
3. **Average Return**
   - This amount is the required return or return requirement.
     - Requirements are more stringent than desires because investors with requirements typically must achieve those returns, at least on average.
4. **Specific return objectives**
   - Should be consistent with that investor's risk objective.
   - Incorporates the required return, the stated return desire, and the risk objective into a measurable annual total return specification.
   - eg investor with 5% after-tax, required, inflation-adjusted annual return but above-average risk tolerance might reasonably set higher

Variation in return requirement & risk tolerance across investors

not that NB

[redacted] Return Requirements and Risk Tolerances of Various Investors

Type of Investor	Return Requirement	Risk Tolerance
Individual	Depends on stage of life, circumstances, and obligations	Varies
Pension Plans (Defined Benefit)	The return that will adequately fund liabilities on an inflation-adjusted basis	Depends on plan and sponsor characteristics, plan features, funding status, and workforce characteristics
Pension Plans (Defined Contribution)	Depends on stage of life of individual participants	Varies with the risk tolerance of individual participants
Foundations and Endowments	The return that will cover annual spending, investment expenses, and expected inflation	Determined by amount of assets relative to needs, but generally above-average or average
Life Insurance Companies	Determined by rates used to determine policyholder reserves	Below average due to factors such as regulatory constraints
Non-Life-Insurance Companies	Determined by the need to price policies competitively and by financial needs	Below average due to factors such as regulatory constraints
Banks	Determined by cost of funds	Varies

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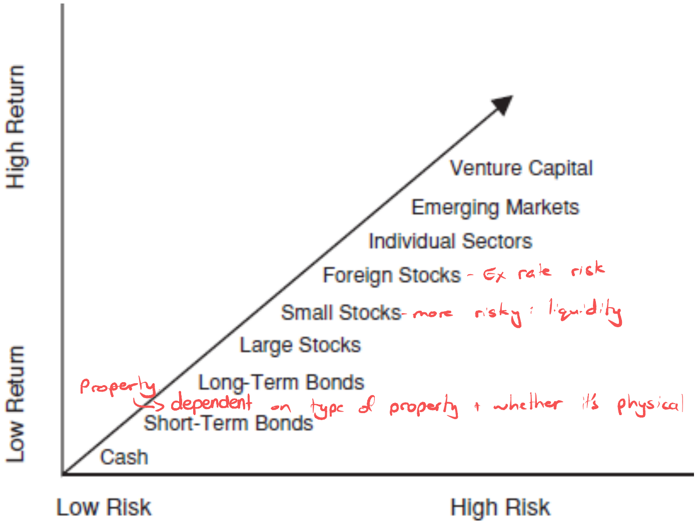
Risk-return by Asset Class



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RSA Asset Classes Returns

- According to South African Saving Institute,over time, the four main asset classes are likely to produce the following real (after inflation) returns in the long run:
  - Cash: 0 to 1%
  - Bonds: 1 to 3%
  - Property: 2 to 4%
  - Equities: 7 to 9%

1:08:40

Investment Restrictions

- Compilation of any financial instruments (e.g., options) or investment strategies (e.g., currency hedging) that are not allowed.
- Can cover entire asset classes, specific transactions, countries, or exchanges, or can be taken to the individual company or organization level
- Often include restrictions based on social, political, or religious reasons.
  - For example a Christian foundation, may wish to avoid investing in "sin" stocks (typically defined as companies that are involved in the manufacture, sale, or distribution of alcohol, tobacco, and firearms).

Investment Constraints

- **Liquidity Constraints**
  - Vary between investors depending on age, employment, tax status, etc.
  - e.g planned vacation expenses and house down payment.
- **Time Constraints**
  - Influences liquidity needs and risk tolerance
  - Longer investment horizons generally require less liquidity and more risk tolerance
  - Two general time horizons are pre-retirement and post-retirement periods

1:10:55

Investment Constraints: Taxes and Interest Income

- Taxes
  - **Unrealized capital gains:** Reflect price appreciation of currently held assets that have not yet been sold
  - **Realized capital gains:** When the asset has been sold at a profit
- Tax Free Investments
  - Earn income that is NOT subject to income taxes, e.g. Tax Free Savings Accounts (TSFA)
- Tax deferred investments
  - Compound tax free but when withdrawn are subject to taxes

1:12:20

Legal and Regulatory Constraints

- Limitations or penalties on withdrawals
- Investment laws prohibit insider trading
- Institutional investors deserve special attentions since legal and regulatory factors may affect them quite differently
- For example amendment to Regulation 28, Pension Fund Act
  - Seeks to ensure that individuals' savings are invested in a sensible way and protected from poorly diversified portfolios
  - Prescribes the maximum exposure that retirement fund investments may have to various asset classes, for example:
    - 25% in property
    - 45% in offshore assets
    - 75% in equities

Personal Constraints: Unique Needs & Preferences

- Personal preferences such as socially conscious investments could influence investment choice
- Time constraints or lack of expertise for managing the portfolio may require professional management
- Large investment in employer's stock may require consideration of diversification needs
- Institutional investor's needs

Typical investment management clients and characteristics

Investor Type	Risk Tolerance	Investment Horizon	Liquidity Needs	Income Needs
Individuals	Depends on Individual	Depends on Individual	Depends on Individual	Depends on Individual
Banks	Low	Short	High	Pay interest
Endowments	High	Long	Low	Spending level
Insurance	Low	Long-life Short-P& C	High	Low
Mutual funds	Depends on fund	Depends on fund	High	Depends on fund
Def'd-benefit pensions	High	Long	Low	Depends on age

1:13:00

Five Common Mistakes Made in IPS

- Miscalculating investor's level of risk tolerance
- Underestimating investment time horizon
- Not determining or inaccurately determining investor's specific financial goals
- Misjudging investor's liquidity needs during their time horizon
- Not identifying investor's unique circumstances or preferences

The future of Portfolio Management

- Becoming a more science-based discipline- technology, and market structure constantly translate into improvements in products and professional practices.
- Incorporation of risk characteristics of nontradable assets in client's portfolio , eg. future earnings from a job, a business, or an expected inheritance,
- For institutional investors, there is an increasing awareness and use of multifactor risk models and methods of managing risk.
- Emergence of a ange of new standardized derivative contracts-swaps, futures, options and crypto-currencies, creating an infinite variety of investment products.

1:13:40

The Ethical Responsibilities Of Portfolio Managers

Don't need to worry about this

- Investment managers should be professional
- Professional standards are of two types:
  - standards of competence
  - standards of conduct
- Ethical conduct is the foundation requirement for managing investment portfolios.

1:18:15 ~ look @ L3 CFA Notes, @ 16-32 for meaning in context

Typical Examination Question

1:20:26 pg 45 solutions @ end

Ms Mandak is 65 years of age, is in excellent health, pursues a simple but active lifestyle, and has one child. She has interest in a private company for R900 million and has decided that, upon her death, 30% of her investment will be donated to an orphanage centre in Cape Town and the rest to her son. She now understands that she needs an appropriate investment policy and asset allocations in order to meet her goals. She currently, has the following assets

- R630 million cash (from sale of the private company interest, net of a R270 million gift to the orphange)
- R100 million stocks and bonds (50 million each)
- R90 million industrial property (currently fully leased) ∴ has income
- R10 million value of her residence
- R650 million total available assets

i. Formulate and justify an investment policy statement setting forth the appropriate guidelines within which future investment actions should take place. Your policy statement should encompass all relevant objectives and constraint considerations.
ii. Recommend and justify a long-term asset allocation that is consistent with the investment policy statement you created in *(i)* above. Briefly explain the key assumptions you made in generating your allocation.

high ability to take risk
- high wealth

low willingness to take risk
- only 50m in stocks,
a lot in cash

Not donating now for tax
benefit upon death